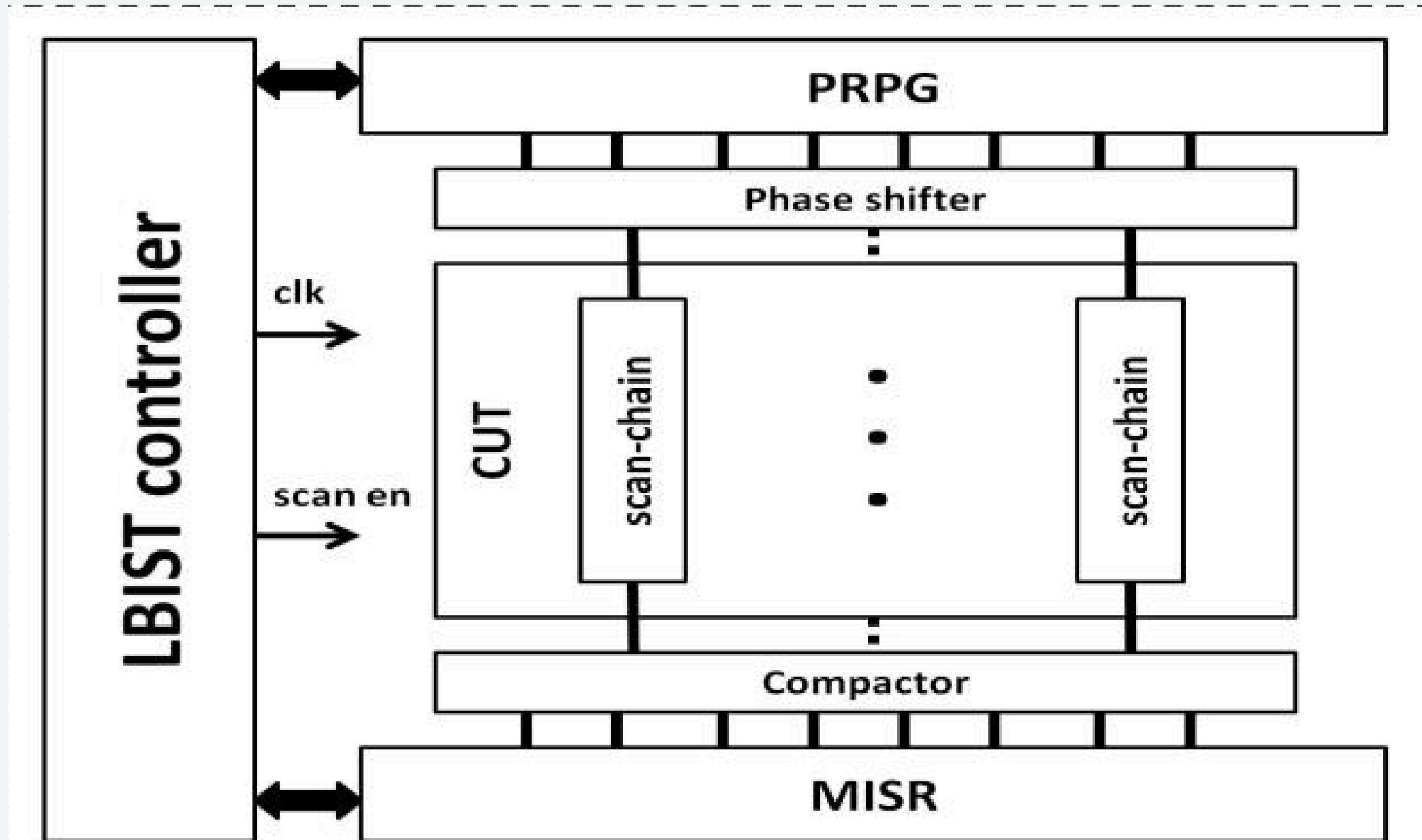


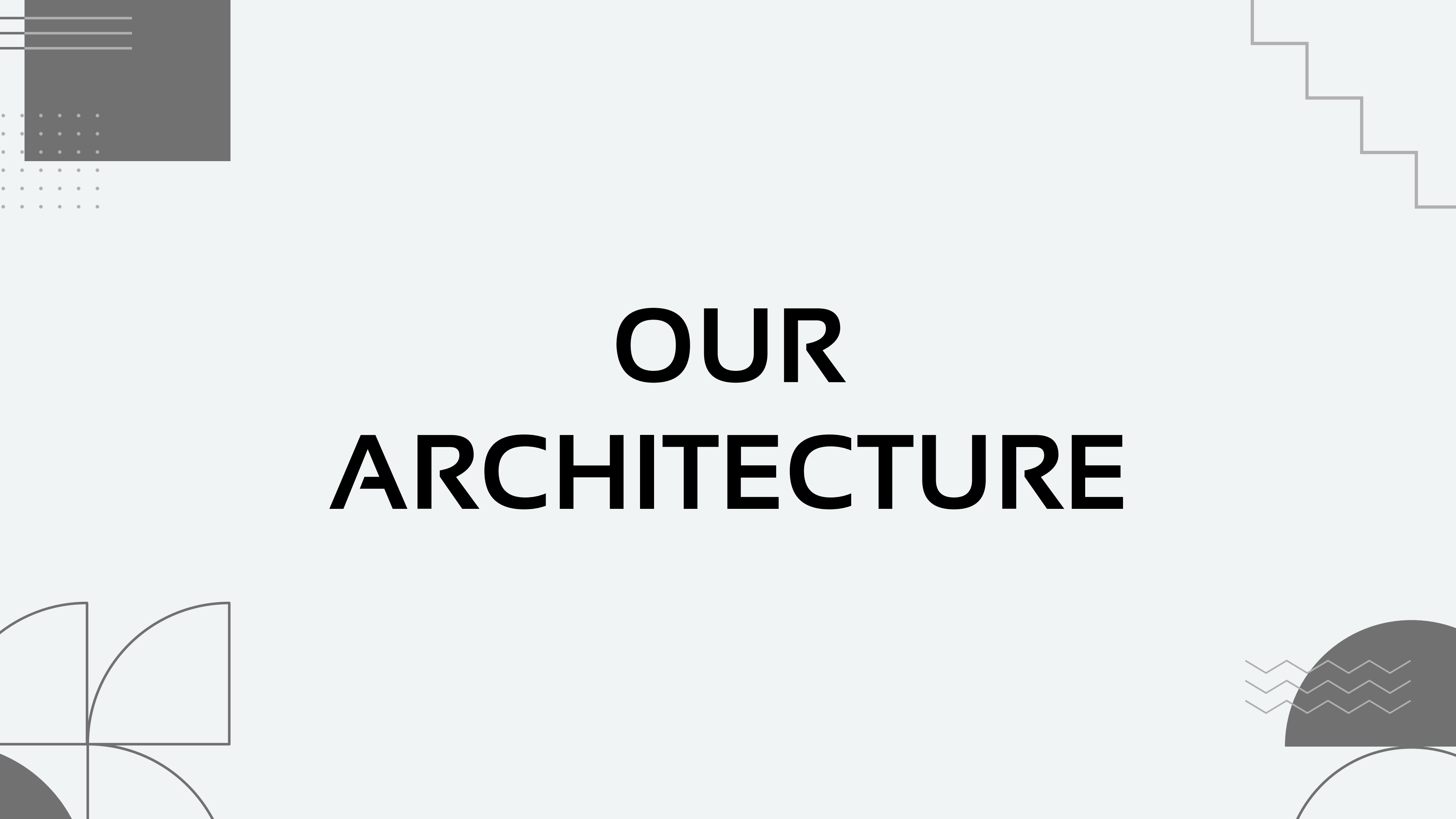
Low Power LBIST For MIPS32

Rohak Gupta (B23500)	Vanilla LBIST implementation and power analysis
Rushabh Lodha (B23495)	PLPF (PLPF2/PLPF3) + masking design and integration in LBIST
Taranpreet Singh (B23504)	LBIST integration in MIPS-32 and final simulation

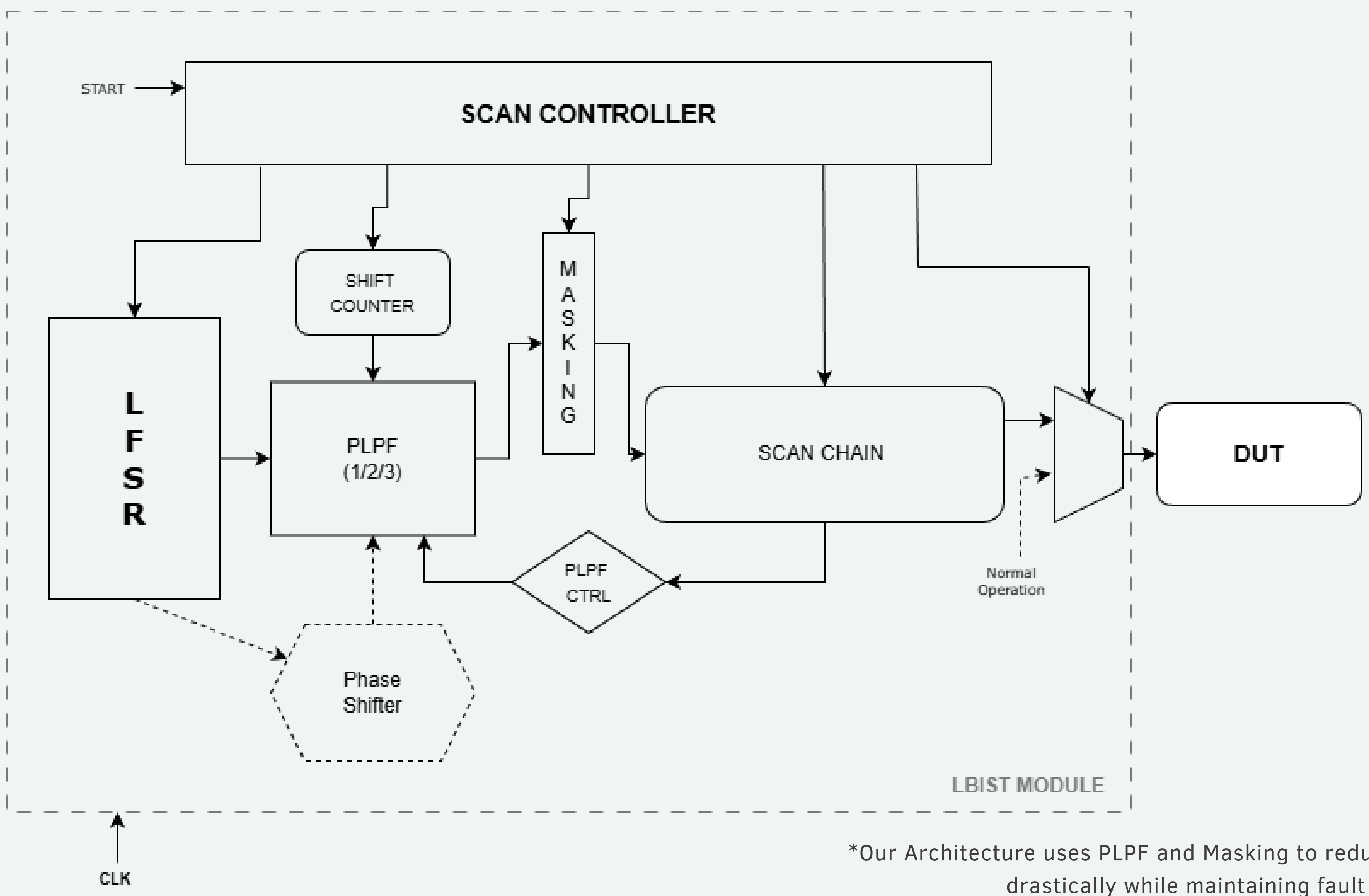


Basic LBIST Architecture

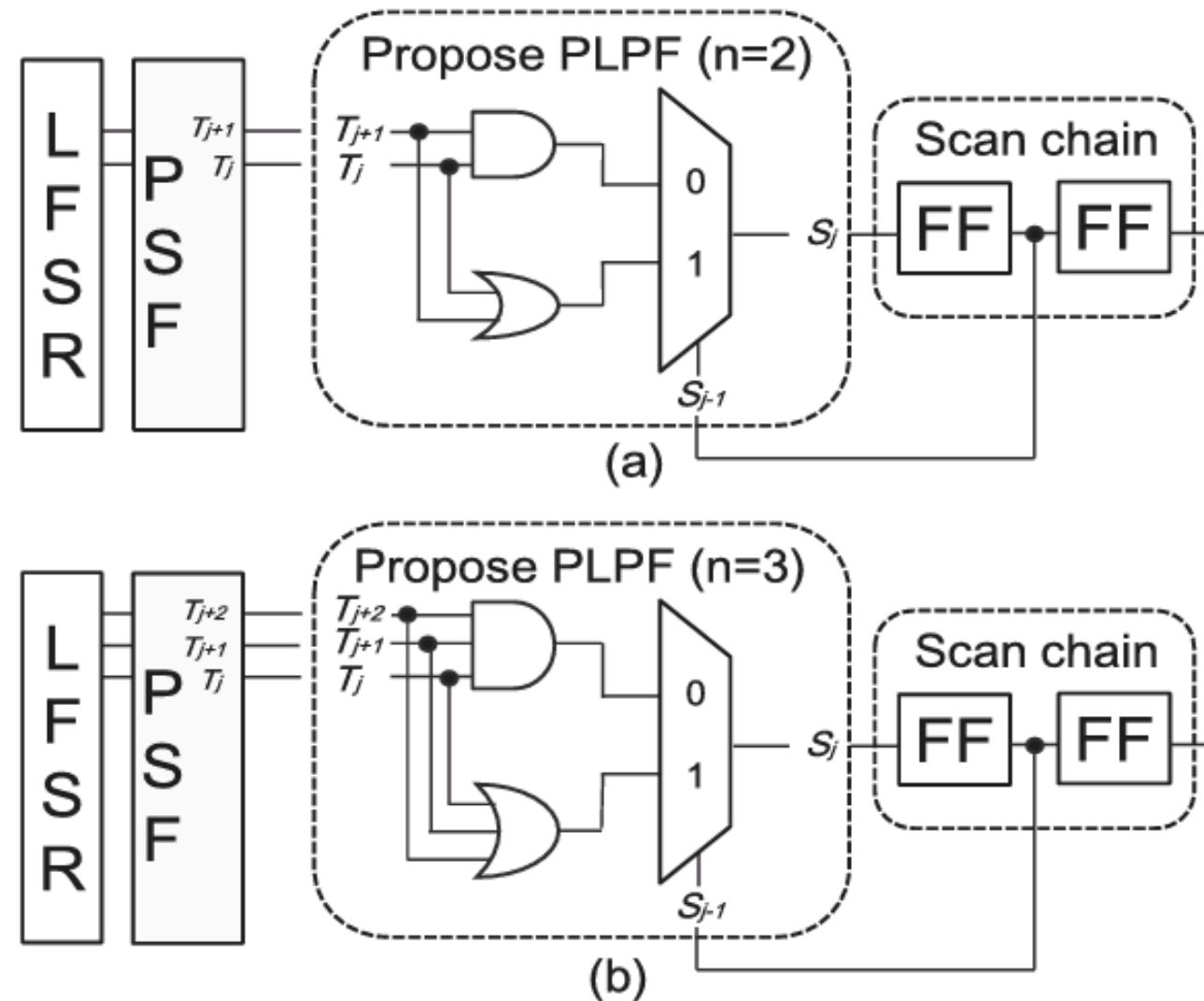




OUR ARCHITECTURE



PLPF and Masking



Optimized PLPFs (a) $n = 2$ (b) $n = 3$.

For a filter span of n bits, PLPF computes:

$$\text{AND-path} = \bigwedge_{i=0}^{n-1} x[n+i], \quad \text{OR-path} = \bigvee_{i=0}^{n-1} x[n+i],$$

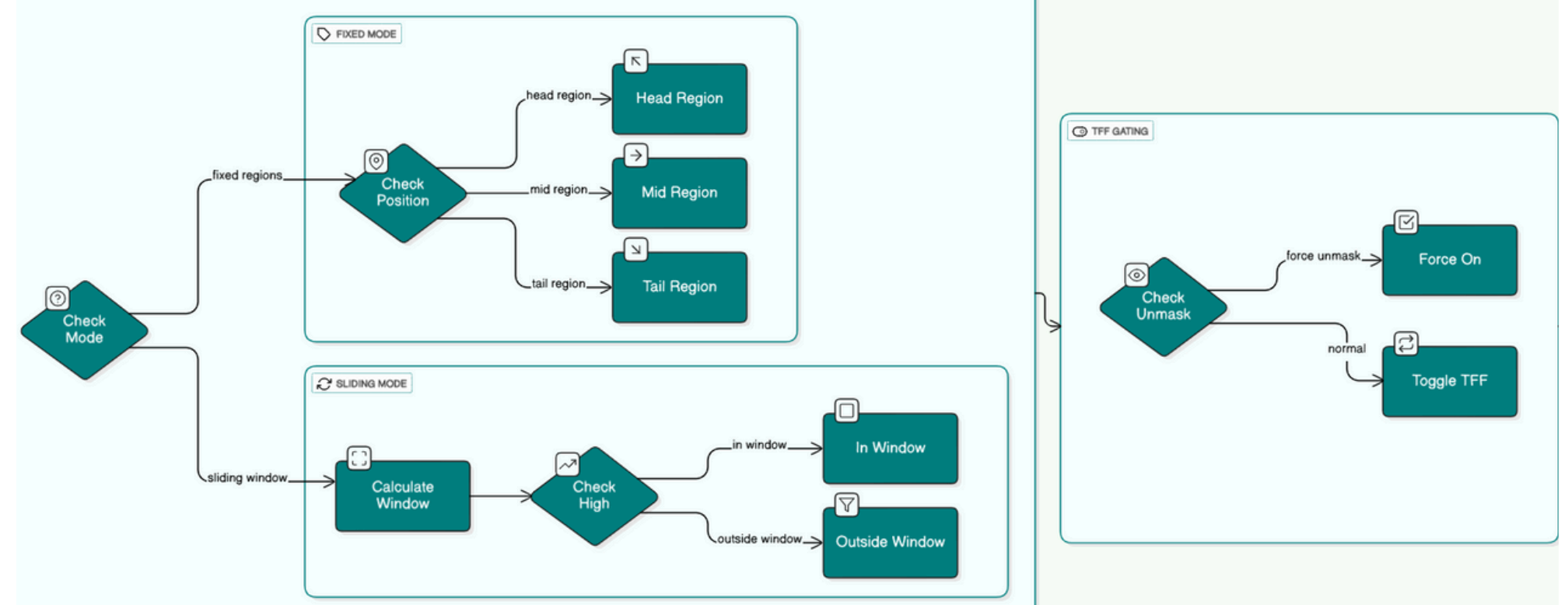
and selects the output according to:

$$b[n] = \begin{cases} \text{AND-path}, & b[n-1] = 0, \\ \text{OR-path}, & b[n-1] = 1. \end{cases}$$



$$T_n \approx \frac{1}{2^{n+1} - 2}.$$

Approximate Toggle Rate (ref: Kato et al.)



Masking periodically bypasses PLPF and re-injects raw LFSR patterns:

$$b[n] = x[n] \quad (\text{every } M\text{-th pattern}).$$

Without masking, the fault activation probability after k filtered patterns is:

$$P_{\text{PLPF}} = 1 - (1 - q)^k.$$

With masking (every M -th pattern unmasked), we have approximately

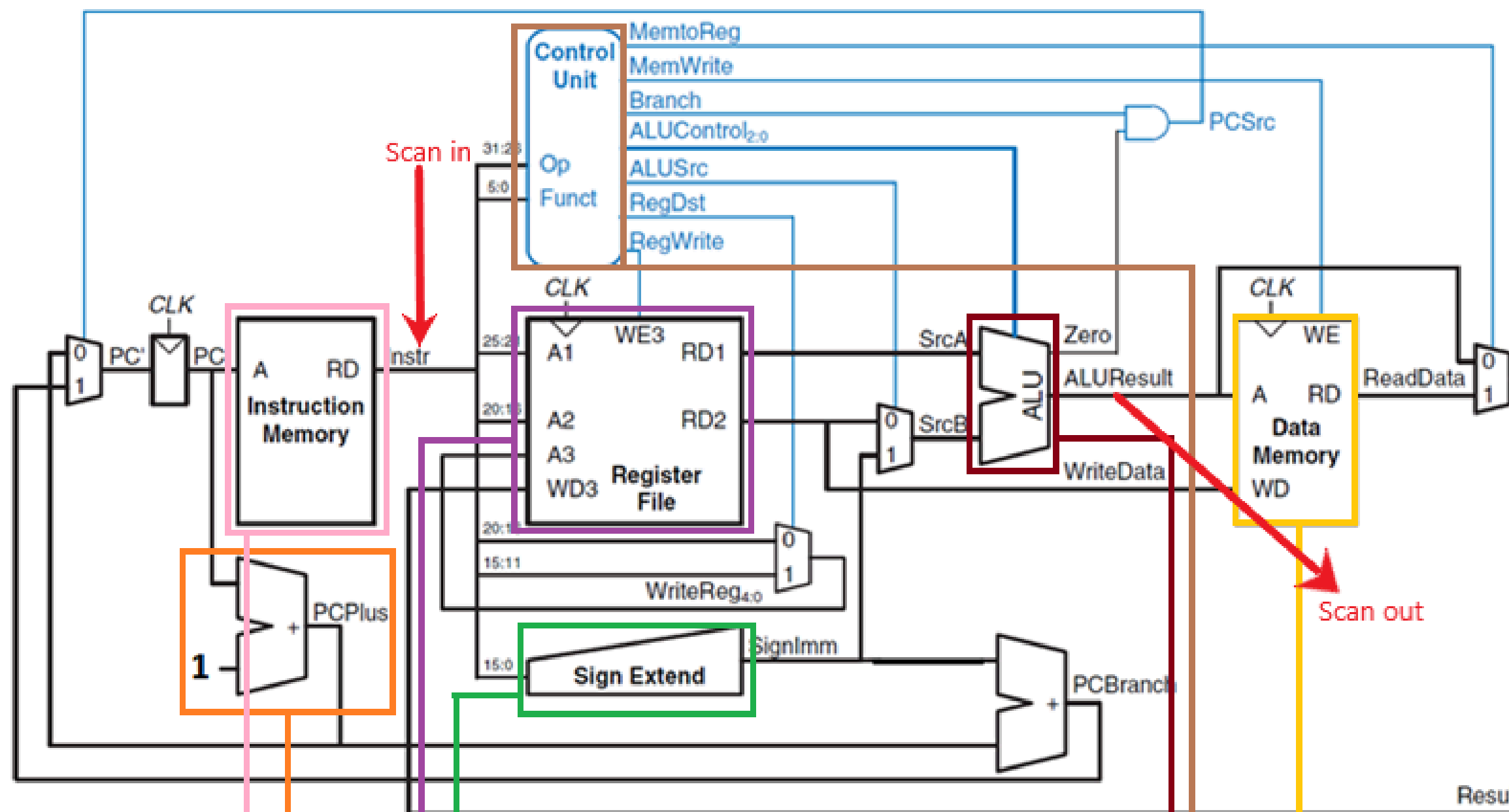
$$P_{\text{masked}} = 1 - (1 - q)^{k - \frac{k}{M}} (1 - p)^{\frac{k}{M}}.$$

- p = probability that a raw LFSR pattern activates a given fault,
- q = activation probability under PLPF (lower, since toggles are filtered),
- M = unmasking period,
- k = total number of applied patterns.



MIPS 32

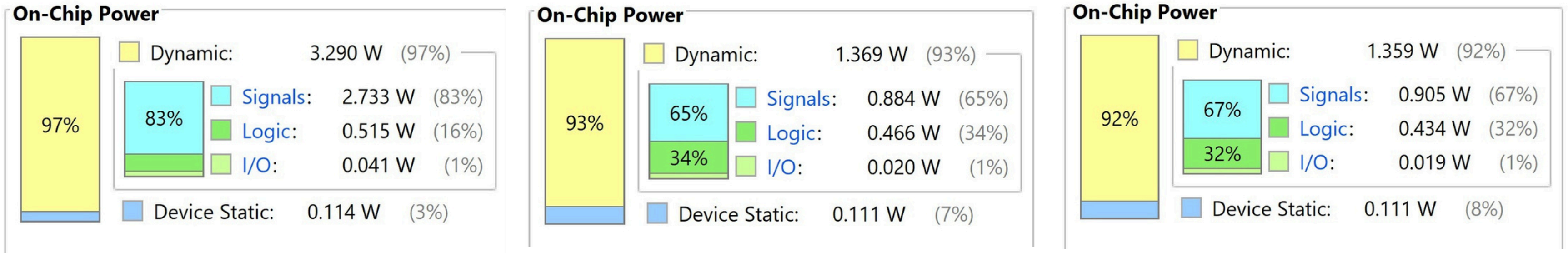
as DUT



Source-Digital Design and Computer Architecture, 2nd Edition form David Harris, Sarah Harris.

1. programcounter.vhd
2. instructionmemory.vhd
3. registerfile.vhd
4. signextend.vhd
5. alu.vhd
6. datamemory.vhd
7. controlunit.vhd

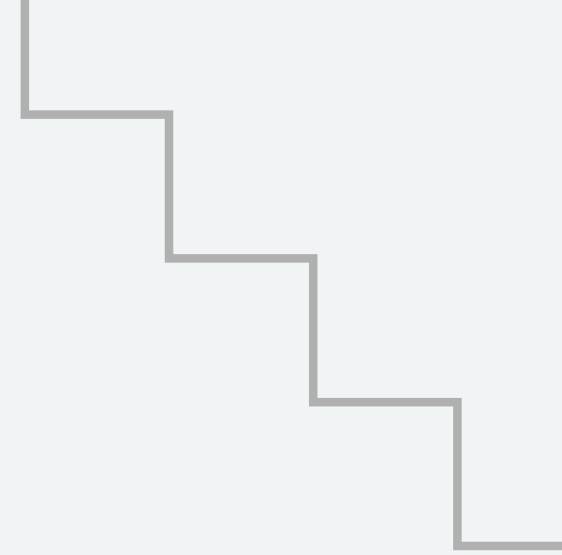
RESULTS



~60% reduction in Power!

Method	Dynamic Power (ALU)	Relative Power	Fault Coverage
Vanilla LBIST (Raw LFSR)	3.290 W	High	High
PLPF Only	1.359 W	Low	Lower
PLPF + Masking	1.369 W	Low (slightly higher)	Slightly Higher

*Synthesis and Implementation were carried out with ALU as our DUT and power was calculated using SAIF files



QNA

